GST Invoice Digitization and Automated Register Generation Mobile Application

Project Synopsis Submitted

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Objectives

- To design and develop an Android application for GST invoice management
- To reduce the manual effort involved in GST record maintenance
- To assist users with the OCR-based extraction of invoice details
- To store GST invoices in a structured and searchable format
- To generate GST registers compatible with Excel for filing and sharing

Problem Statement

Small businesses, startups, and individuals are required to maintain structured GST records for tax filing and compliance. However, invoices are often received as unstructured documents such as PDFs, scanned copies, or mobile photographs. Manually extracting invoice details and entering them into Excel sheets is time-consuming, error-prone, and inefficient. Small businesses, startups, and individuals often manually read these invoices and enter details into Excel sheets for record keeping and filing, which is time-consuming, repetitive, and prone to human error. There is a need for a mobile-based solution that assists users in capturing invoice data efficiently and generating structured GST registers.

Proposed Solution

The proposed system is an Android mobile application that assists users in managing GST invoices by combining OCR-assisted data extraction with structured manual verification. The application allows users to upload GST invoice images or PDF files, automatically extract key textual information using OCR, and present the extracted data as editable form fields. After user verification, the application stores invoice records locally, can search and categorize them, and generates GST-compliant registers in Excel-compatible formats for easy sharing and filing.

Scope of the Project

The scope of the project includes:

- Uploading GST invoices as images or PDF documents
- OCR-based text extraction to assist data entry

- Manual verification and correction of extracted invoice details
- Local storage of invoice records using a database
- Exporting invoice data as CSV / Excel-compatible files

The project does not aim to fully automate GST filing or replace accounting software. It focuses on invoice digitization and register generation only.

System Features

- **Invoice Upload & Preview:** Users can upload or capture GST invoice documents and preview them within the app.
- **OCR-Assisted Data Extraction:** OCR is used to detect invoice text such as GSTIN, invoice number, date, and amounts.
- **Structured Data Entry & Validation:** Extracted data is shown in editable form fields with validation rules.
- **Invoice Storage:** Verified invoice records are stored locally for future access.
- **Search & Filters:** Users can search invoices by vendor, invoice number, or date range.
- **GST Register Export:** The app generates structured GST registers in Excel/CSV compatible formats.
- **Share Functionality:** Generated files can be shared as Excel/CSV via email, messaging apps, or cloud storage.

Technology Stack

- **Platform:** Android
- **Programming Language:** Java / Kotlin
- **Database:** Room (SQLite)
- **OCR Engine:** ML Kit Text Recognition/ Tesserect OCR

- **UI Design:** XML layouts with Material Components
- **File Handling:** Storage Access Framework
- **Export Formats:** CSV / Excel-compatible formats

Use Cases

- Small businesses maintaining monthly GST purchase registers
- Freelancers and consultants tracking GST invoices
- Startups sharing GST expense data with accountants
- Individuals organizing reimbursement or expense invoices

Limitations

- OCR accuracy depends on invoice quality and formatting
- Manual verification is required to ensure correctness
- Invoice layouts vary across vendors
- The system does not perform GST return filing

Future Enhancements

- Improved OCR-based field detection using layout analysis
- Cloud backup and multi-device synchronization
- Monthly GST summary analytics
- Integration with accounting and GST filing platforms

Conclusion

The GST Invoice Digitization application provides a solution for managing GST invoice records using a mobile platform. By combining OCR-assisted extraction with user verification and structured export, the system reduces manual effort while maintaining accuracy.

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I. Literature Review

Optical Character Recognition (OCR) serves as a key technology for automatically extracting structured information from unstructured invoice images and PDFs, especially in areas such as GST compliance reporting in India.Small businesses and startups continue to struggle with the manual entry of invoice details such as GSTIN, invoice number, date, taxable value, tax amounts, and totals into registers or filing systems. Existing literature demonstrates that hybrid pipelines combining image preprocessing, deep learning-based detection, and rule-based post-processing can achieve high accuracy while reducing manual effort. However, most prior systems remain desktop-oriented, cloud-dependent, or lack end-to-end register generation and offline-first mobile support tailored to Indian GST formats.

A. Offline and Traditional OCR Methods for Scanned Documents

These methods rely on straightforward image-processing steps such as skew correction, noise removal, binarization, and line/character segmentation, and are best suited for clean scanned inputs.

Majumder et al. (2019) [1] proposed a complete offline OCR system for scanned English documents (primarily Calibri font). It includes skew correction, noise removal, binarization, line/character segmentation, and correlation-based recognition. Tested on 50 images, it achieved 92% average accuracy and 98% for Calibri font, with low computational time. The approach works reliably on well-aligned scans but assumes consistent fonts and clean inputs, which does not hold for distorted or noisy mobile-captured invoices.

B. Mobile/Android-Based OCR for Receipts and Bills

Mobile-oriented systems generally focus on preprocessing and OCR tailored for images captured by smartphones, with performance often tested under realistic conditions such as varying distance and image quality.

Rahmat et al. (2018) [2] developed an Android application using Convolutional Neural Networks (CNN) for receipt text recognition. The system applies multiple preprocessing steps: grayscale conversion, sharpening, denoising, adaptive thresholding, and morphological operations before recognition. Extracted data is stored in a database for tax sampling. Accuracy reached 95% when images were taken at 10 cm and 85% at 12 cm for key fields such as subtotal and tax. This highlights mobile feasibility but is limited to simple receipt layouts without GST-specific fields or register export.

Kumar et al. (2020) [3] proposed an offline Android pipeline combining OpenCV preprocessing with Tesseract OCR. The method emphasizes shadow and watermark removal through morphological operations, adaptive thresholding, and median blur, along with bifurcation to manage long invoices. It achieved 97% accuracy on small bills and 83% on larger ones, reliably extracting total amount and date. However, it requires manual alignment and lacks GST field mapping, tax logic validation, or structured register output.

C. Deep Learning and Integrated/Region-Aware OCR Systems

More recent systems combine document classification, stronger preprocessing, and region-aware detection to handle a wider variety of real-world documents.

Mokdam et al. (2025) [4] built an integrated pipeline that first classifies documents (scientific papers, letters, forms, memoranda) using a CNN, then applies CLAHE, adaptive thresholding, sharpening, erosion, morphological operations, and contour analysis to identify text and tables, before using region-specific Tesseract OCR. Tested on 200 varied scanned documents, it achieved 69% end-to-end accuracy (65.4% precision, 66.7% recall), with preprocessing noticeably improving results. A Tkinter GUI supports interactive viewing and structured export. While the classification step is valuable, overall accuracy remains modest on complex layouts, and the desktop focus limits mobile GST applications.

Gawade et al. (2025) [5] presented a hybrid approach that uses OCR (Tesseract, AWS Textract, Google Cloud Vision) to extract raw text from invoices, followed by an LLM (fine-tuned BERT or GPT-4) for named-entity recognition, field mapping, validation, and error correction. The system manages diverse layouts, reduces manual effort, and improves accuracy by verifying field relationships (e.g., line items vs. total). It speeds up processing and supports ERP integration, though its reliance on cloud-based LLMs makes it computationally heavy and less suitable for low-resource Indian MSME settings.

Arslan (2022) [6] eveloped an end-to-end invoice processing system accepting submissions via web or email. Text files use template matching; images undergo morphological preprocessing and YOLOv5 table detection, cell extraction, and OCR with a custom Tesseract model fine-tuned on 957 Turkish invoices (TTUR). This reduced character error rate from 11% to 6.07%, with YOLOv5 reaching 98.07% mAP@0.5 for tables. The work shows strong results from domain-specific fine-tuning and hybrid methods, but the web/email focus is less practical for small businesses relying on mobile photos.

D. Multi-Engine Comparison and Structured Output Generation

These works compare OCR engines and produce structured formats (JSON/CSV) after preprocessing.

Kamisetty et al. (2022) [7] built an Android app evaluating Tesseract, EasyOCR, and Keras-OCR after OpenCV preprocessing (grayscale, inverted, noise removal, canny, thick font). Tesseract performed best; cleaned text was output as JSON/CSV with app-based input/storage/viewing. This demonstrates offline multi-engine comparison and structured export but lacks GST-specific validation, editable forms, searchable local database, or compliance register generation.

Singh et al. (2024) [8] presented a comprehensive survey of advancements and challenges in OCR-based invoice data extraction systems (Invoice Recognition Systems – IRS). The study highlights the evolution from rule-based methods to deep learning pipelines (YOLO/EAST for detection, Tesseract/EasyOCR for recognition) and emphasizes hybrid approaches with preprocessing, region-aware extraction, and post-processing (regex, spell-checking) as currently achieving the highest real-world accuracy. It underscores the need for domain-specific adaptation and end-to-end automation, aligning closely with the requirements of GST compliance applications.

E. Human-in-the-Loop Approaches

Hybrid systems that integrate automated extraction with user intervention offer a practical means to address errors inherent in real-world document images, where fully autonomous OCR often falls short.

Jung et al. (2021) [9] proposed a mixed-initiative pipeline for medical invoices using CNN table detection, heuristics, and interactive user corrections via visualizations. It achieved $\sim$97% relation level accuracy with low user effort ($\sim$80 seconds per invoice). This validates human-in-the-loop verification for variable layouts highly relevant to ensuring GST data accuracy through mandatory editing.

F. Surveys and Analytical Studies on Invoice OCR

Several survey papers and application-oriented studies offer valuable perspectives on techniques for text extraction from invoices and bills, while also revealing persistent challenges in achieving compliance ready outputs.

Yindumathi et al. (2020) [10] examined a range of machine learning and OCR techniques applied to invoice and bill documents. Their analysis showed that well-designed preprocessing pipelines, incorporating thresholding, noise suppression, and edge detection paired with hybrid recognition engines deliver substantially better performance than standalone OCR tools. Nonetheless, the majority of approaches reviewed remain focused on raw text extraction and seldom advance to structured, regulation compliant formats such as GST compliant registers.

Reddy and Krishna (2025) [11] presented a mobile application tailored for inventory and billing management among Indian micro, small, and medium enterprises (MSMEs). The solution incorporates real time stock monitoring, barcode enabled billing, automated low-stock notifications, UPI payment support, and convenient sharing of digital receipts through WhatsApp, email, or SMS. Although the primary focus lies on invoice creation and inventory control rather than retrospective data extraction, the work effectively illustrates the strong and growing demand for offline capable, mobile-centric tools in the Indian small-business ecosystem. This emphasis on generation rather than extraction precisely underscores the market opportunity that the present GST digitization and register generation system.

Andriansah and Kristiyanto (2025) [12] developed a web-based platform for tax invoice recapitulation in an Indonesian enterprise context, employing OCR to automatically derive structured data from both PDF and JPG invoice files. The application attained 100% accuracy on PDF inputs and 89% on JPGs, while reducing processing time by 92.92% compared to manual Excel-based entry. This demonstrates the viability of OCR-supported tax compliance workflows in a comparable regulatory environment. However, the system is entirely server-dependent, offers no offline functionality, and lacks adaptation to GST specific requirements. The marked decline in performance from PDFs to JPGs a situation routinely encountered when invoices are photographed with smartphones—further reinforces the necessity of robust on-device preprocessing and enforced user validation, both central to the proposed design.

G. Gaps in Existing Literature

Collectively, the reviewed body of work establishes a solid technical foundation encompassing effective image preprocessing, early mobile OCR implementations, comparative evaluations of recognition engines, hybrid deep-learning pipelines, domain-adapted model tuning, user-assisted correction strategies, and selected applications in tax-related data handling.

Nevertheless, important gaps remain when addressing the specific needs of Indian GST compliance, particularly for small-scale taxpayers, freelancers, and startups operating with limited resources:

- Limited attention to GST domain-specific extraction and validation requirements, including verification of 15-character GSTIN formats, accurate classification of IGST versus CGST + SGST components, and enforcement of proper rounding rules.
- Lack of mechanisms for automatic generation of structured GST registers (purchase and sales ledgers) in Excel or CSV compatible formats that include all mandatory compliance fields.
- Lack of fully offline mobile solutions that incorporate local searchable databases (such as Room with SQLite), editable verification interfaces, historical record browsing, filtering options, and seamless sharing capabilities.
- Insufficient emphasis on mandatory user verification to guarantee 100% correctness given OCR's sensitivity to image quality/layout variation.

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TABLE I
COMPARISON OF GST INVOICE DIGITIZATION AND AUTOMATED REGISTER GENERATION SOLUTIONS

Reference	**Year**	**Platform**	**Offline OCR**	**Preprocessing**	**OCR Engine**	**GST-specific**	**Editable Verification**	**Local DB**	**GST Register**	**Key Features/Limitations**
Majumder et al.	2019	Desktop	Yes	Skew correction, noise removal, binarization, segmentation	Correlation-based	No	No	No	No	92% acc on clean scans; struggles with mobile distortion, mixed fonts, noise
Rahmat et al.	2018	Android	Partial	Grayscale, sharpening, denoising, adaptive thresh, morphological	CNN	Partial	Partial	Partial	No	95% @10cm / 85% @12cm; simple receipts; no GST fields/register export
Kumar et al.	2020	Android	Yes	Morphological ops, adaptive thresh, median blur, shadow removal	OpenCV + Tesseract	Partial	No	No	No	97% small / 83% large bills; manual alignment; no GST validation/register
Mokdam et al.	2025	Desktop	No	CLAHE, adaptive thresh, sharpening, erosion, contour	CNN + region-aware Tesseract	No	Partial	No	No	69% end-to-end; modest acc; not mobile/GST-focused
Kamisetty et al.	2022	Android	Yes	Grayscale, invert, noise removal, Canny	Compares Tesseract, EasyOCR, Keras-OCR	No	No	No	Partial	JSON/CSV export; lacks GST validation/editable forms/DB
Arslan	2022	Web	No	Noise reduction, morphological	Tesseract fine-tuned Turkish	Partial	No	No	No	YOLOv5 table detection; Turkish invoices; no GST
Jung et al.	2021	Desktop/web	No	CNN table detection + heuristics	Not primary focus	No	Yes	No	No	∼97% acc; strong human-in-the-loop; not mobile/GST
Reddy and Krishna	2025	Android	Partial	No	No OCR	Partial	Partial	Yes	Partial	MSME billing/inventory, barcode, UPI; no OCR/GST register
Andriansah and Kristiyanto	2025	Web	No	Not detailed	Tesseract + PDFplumber	Partial	No	Partial	Partial	Indonesian tax recapitulation; 100% PDF acc; server-based
Vyapar	Ongoing	Android	Partial	Server-side	Proprietary AI/cloud OCR	Partial	Partial	Yes	Partial	99.9% claimed acc; cloud OCR; offline billing; GSTR reports
GimBooks	Ongoing	Android	Partial	Server-side	Proprietary	Partial	Partial	Yes	Partial	GST billing, e-invoice, inventory; cloud OCR; GSTR export